Exploratory Data Analysis using Visualization

## 1. Problem Statement & Objective

This report analyses passenger data from the Titanic. The dataset holds details on passenger class, age, gender, fare, family members, ticket and survival status. The objective is to perform Exploratory Data Analysis (EDA) with Matplotlib and Seaborn to understand the passenger distribution, the survival patterns, and the relationships between different variables, and then to draw insights from the visualizations.

| Column | Description |
|---|---|
| PassengerId | Unique ID of each passenger |
| Survived | Survival status (0 = No, 1 = Yes) |
| Pclass | Passenger class (1 = First, 2 = Second, 3 = Third) |
| Name / Sex / Age | Passenger name, gender and age in years |
| SibSp | Number of siblings / spouses aboard |
| Parch | Number of parents / children aboard |
| Ticket / Fare / Cabin | Ticket number, fare paid, cabin number |
| Embarked | Port of embarkation (C = Cherbourg, Q = Queenstown, S = Southampton) |

## 2. Importing Libraries & Loading the Dataset

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style('whitegrid')

# Load the Titanic dataset
df = pd.read_csv('Titanic.csv')
df.head()
```

The dataset was loaded successfully. It contains 891 rows (passengers) and 12 columns.

## 3. Basic Data Exploration

### 3.1 Dataset Information – df.info()

```
print(df.shape)
```

```
df.info()
```

| Sno. | Column | Non-Null Count | Dtype |
|---|---|---|---|
| 0 | PassengerId | 891 non-null | int64 |
| 1 | Survived | 891 non-null | int64 |
| 2 | Pclass | 891 non-null | int64 |
| 3 | Name | 891 non-null | object |
| 4 | Sex | 891 non-null | object |
| 5 | Age | 714 non-null | float64 |
| 6 | SibSp | 891 non-null | int64 |
| 7 | Parch | 891 non-null | int64 |
| 8 | Ticket | 891 non-null | object |
| 9 | Fare | 891 non-null | float64 |
| 10 | Cabin | 204 non-null | object |
| 11 | Embarked | 889 non-null | object |

Output of df.info() – Shape: (891, 12)

Observation: The dataset has 891 entries. Age (714 non-null), Cabin (204 non-null) and Embarked (889 non-null) contain missing values. The data types are 5 integer, 2 float and 5 object columns.

3.2 Statistical Summary – df.describe()

```
df.describe()
```

| stat | PassengerId | Survived | Pclass | Age | SibSp | Parch | Fare |
|---|---|---|---|---|---|---|---|
| count | 891.00 | 891.00 | 891.00 | 714.00 | 891.00 | 891.00 | 891.00 |
| mean | 446.00 | 0.38 | 2.31 | 29.70 | 0.52 | 0.38 | 32.20 |
| std | 257.35 | 0.49 | 0.84 | 14.53 | 1.10 | 0.81 | 49.69 |
| min | 1.00 | 0.00 | 1.00 | 0.42 | 0.00 | 0.00 | 0.00 |
| 25% | 223.50 | 0.00 | 2.00 | 20.12 | 0.00 | 0.00 | 7.91 |
| 50% | 446.00 | 0.00 | 3.00 | 28.00 | 0.00 | 0.00 | 14.45 |
| 75% | 668.50 | 1.00 | 3.00 | 38.00 | 1.00 | 0.00 | 31.00 |
| max | 891.00 | 1.00 | 3.00 | 80.00 | 8.00 | 6.00 | 512.33 |

Output of df.describe() – numerical columns

Observation: The average age is about 29.7 years (median 28). The average fare is £32.20, but the median fare is only £14.45, and the maximum fare is £512.33. This large gap between mean and median shows that Fare is heavily right-skewed. Overall, 38.4% of passengers survived (mean of Survived = 0.384).

4. Handling Missing Values

```
# Missing values before cleaning
print(df.isnull().sum())

# Age: fill with median (robust to outliers)
df['Age'] = df['Age'].fillna(df['Age'].median())

# Embarked: fill with mode (most frequent port)
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])

# Cabin: ~77% missing, so drop the column
df = df.drop(columns=['Cabin'])

# Missing values after cleaning
print(df.isnull().sum())
```

| Column | Missing (Before) | % Missing | Treatment | Missing (After) |
|---|---|---|---|---|
| Age | 177 | 19.9% | Filled with median (28.0) | 0 |
| Cabin | 687 | 77.1% | Column dropped | – |
| Embarked | 2 | 0.2% | Filled with mode ("S") | 0 |

Reasoning: Age has ~20% missing values, so removing those rows would waste a lot of data; the median is used because age is slightly skewed. Cabin is missing for over three-quarters of passengers, so it carries too little information to impute reliably. Embarked has only 2 missing values, so the most frequent port (Southampton) is a safe replacement.

Feature Engineering

```
# Family size = siblings/spouses + parents/children
df['FamilySize'] = df['SibSp'] + df['Parch']
```

A new column FamilySize was created (SibSp + Parch). It is used in Question 5(b) and the correlation heatmap.

5. Data Visualization & Interpretation

Q1. Create a countplot to show the number of passengers by gender.

```
plt.figure(figsize=(6, 4))
```

```
ax = sns.countplot(x='Sex', data=df, hue='Sex', palette=['#4C72B0', '#DD8452'],
legend=False)
for container in ax.containers:
        ax.bar_label(container)
plt.title('Number of Passengers by Gender')
plt.xlabel('Gender')
plt.ylabel('Number of Passengers')
plt.show()
```

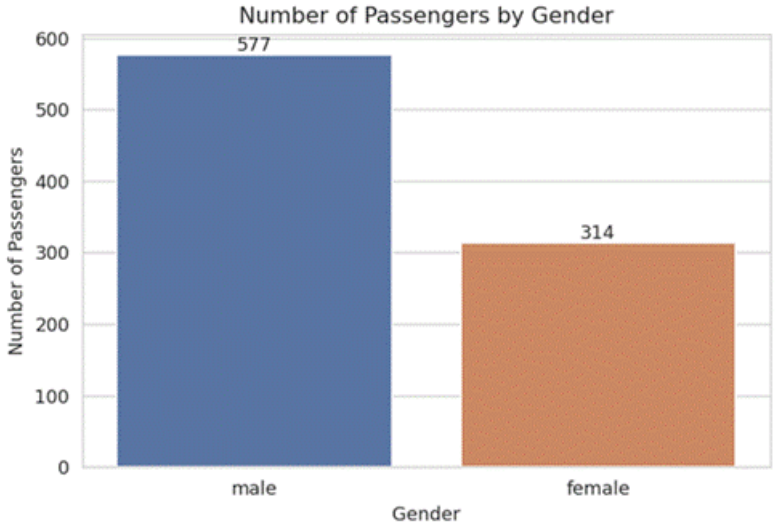



Figure 1: Number of passengers by gender

Interpretation

- There were 577 male passengers (64.8%) and 314 female passengers (35.2%).
- Men outnumbered women by almost 2 to 1 (about 1.8 men for every woman).
- This imbalance matters later: gender turns out to be the strongest factor in survival, so the gender mix strongly shapes the overall survival figures.

Q2. Create a bar chart to show the number of passengers in each passenger class using different colors.

```
class_counts = df['Pclass'].value_counts().sort_index()

plt.figure(figsize=(6, 4))
bars = plt.bar(['1st Class', '2nd Class', '3rd Class'], class_counts.values,
                color=['#55A868', '#C44E52', '#8172B3'])
plt.bar_label(bars)
plt.title('Number of Passengers in Each Passenger Class')
plt.xlabel('Passenger Class')
plt.ylabel('Number of Passengers')
plt.show()
```

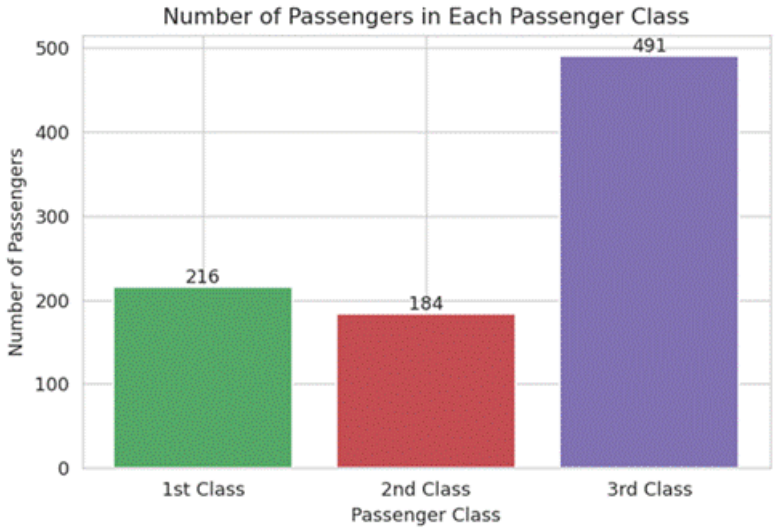



Figure 2: Passengers per class

Interpretation

- 3rd class had the most passengers (491, 55.1%), followed by 1st class (216, 24.2%) and 2nd class (184, 20.7%).
- More than half of everyone on board travelled in the cheapest class.
- First-class passengers were a small minority, yet, as later charts show, they had a clearly better chance of survival.

Q3.Create a histogram to show the distribution of passenger age.
plt.figure(figsize=(7, 4))

```
sns.histplot(df['Age'], bins=30, kde=True, color='#4C72B0')
plt.title('Distribution of Passenger Age')
plt.xlabel('Age (years)')
plt.ylabel('Number of Passengers')
plt.show()
```

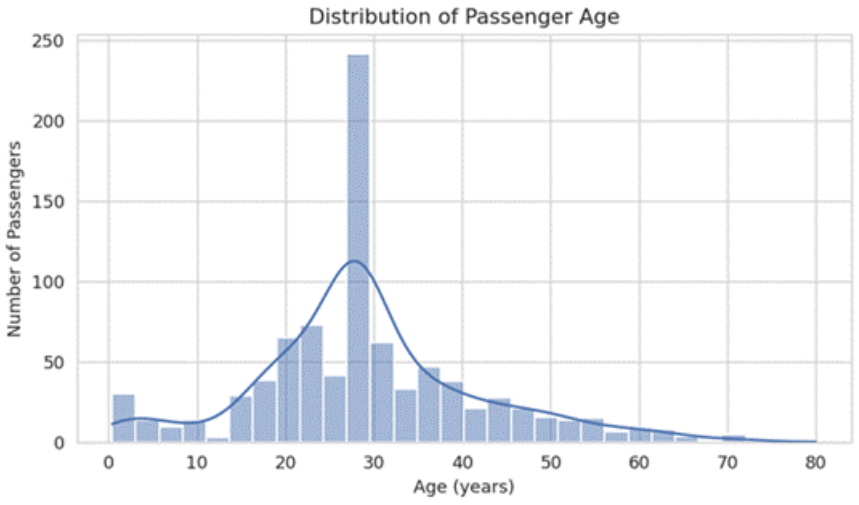


Figure 3: Age distribution (with KDE curve)

Interpretation

- The distribution is roughly bell-shaped and slightly right-skewed (skewness ≈ 0.51), with a tail of older passengers up to 80 years.
- Most passengers were young adults: about 564 passengers were aged 20–39 years, and the mean age is ≈ 29 years.
- There is a small, separate bump at ages 0–5 (infants and toddlers), so a noticeable number of young children were on board. About 69 passengers were aged 12 or under, while only 26 were 60 or older.
- The tall spike near age 28 is caused by our median imputation: all 177 missing ages were filled with 28.

Q4. Create a boxplot to detect outliers in the fare column.

```
plt.figure(figsize=(7, 3.5))
sns.boxplot(x=df['Fare'], color='#8FBC8F')
plt.title('Boxplot of Fare (Outlier Detection)')
plt.xlabel('Fare (£)')
plt.show()

# Outlier count using the IQR rule
Q1, Q3 = df['Fare'].quantile([0.25, 0.75])
IQR = Q3 - Q1
upper_limit = Q3 + 1.5 * IQR
print('Upper limit:', upper_limit)
print('Outliers   :', (df['Fare'] > upper_limit).sum())
```

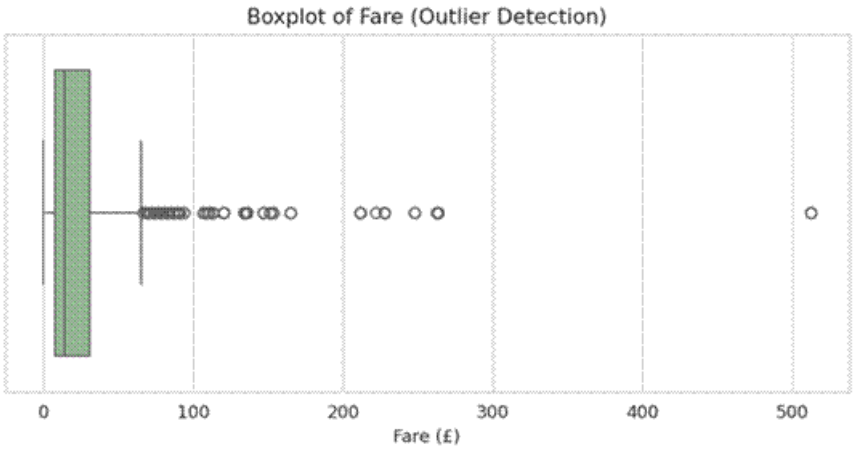



Figure 4: Boxplot of Fare

| Q1 | Median | Q3 | IQR | Upper Limit (Q3 + 1.5×IQR) | Outliers | Max Fare |
|---|---|---|---|---|---|---|
| £7.91 | £14.45 | £31.00 | £23.09 | £65.63 | 116 | £512.33 |

Interpretation

- The box is very compressed near the low end, so most passengers paid modest fares (50% paid between £7.91 and £31.00).
- Many points lie far to the right of the upper whisker: 116 passengers paid more than the IQR limit of £65.63, so they are flagged as outliers.
- The most extreme values are three tickets at £512.33, far above everything else. Also, 15 passengers have a fare of £0 (possibly crew or free tickets).
- These outliers are genuine (expensive first-class suites), not data errors, so they were kept. They do, however, pull the mean fare (£32.20) well above the median (£14.45).

Q5. Create a scatterplot to  a. show the relationship between age and fare. b. analyze the relationship between family size (SibSp + Parch) and fare

(a) Relationship between Age and Fare

```
plt.figure(figsize=(7, 4.5))
sns.scatterplot(x='Age', y='Fare', data=df, alpha=0.6, color='#4C72B0')
plt.title('Age vs Fare')
plt.xlabel('Age (years)')
plt.ylabel('Fare (£)')
plt.show()

print(df[['Age', 'Fare']].corr())
```

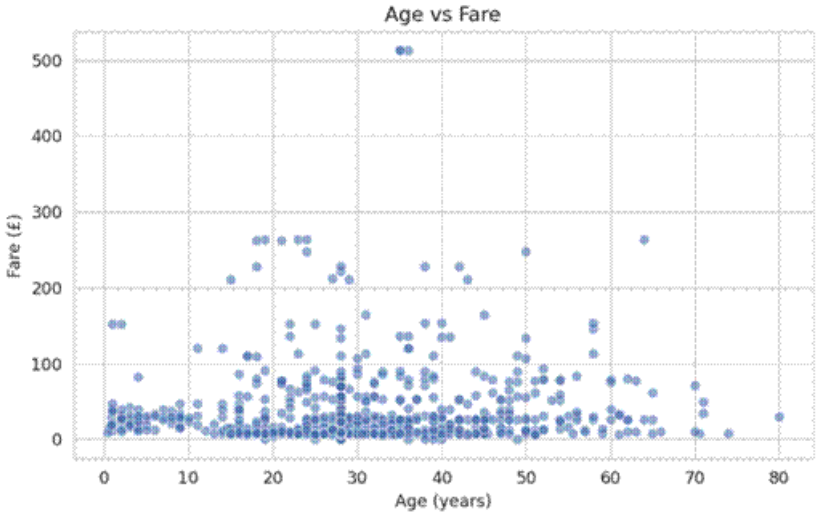



Figure 5(a): Age vs Fare

Interpretation

- There is almost no linear relationship between age and fare: the correlation is only r ≈ 0.10.
- Points are spread across all ages, and low fares dominate at every age. Both young and old passengers paid cheap and expensive fares.

- The 20 passengers who paid above £200 range in age from 15 to 64 (most between 18 and 43), so even very high fares are not tied to any one age group. Age is therefore not a useful predictor of fare.

(b) Relationship between Family Size (SibSp + Parch) and Fare

```
plt.figure(figsize=(7, 4.5))
sns.scatterplot(x='FamilySize', y='Fare', data=df, alpha=0.6, color='#C44E52')
plt.title('Family Size vs Fare')
plt.xlabel('Family Size (SibSp + Parch)')
plt.ylabel('Fare (£)')
plt.show()

print(df[['FamilySize', 'Fare']].corr())
print(df.groupby('FamilySize')['Fare'].mean())
```

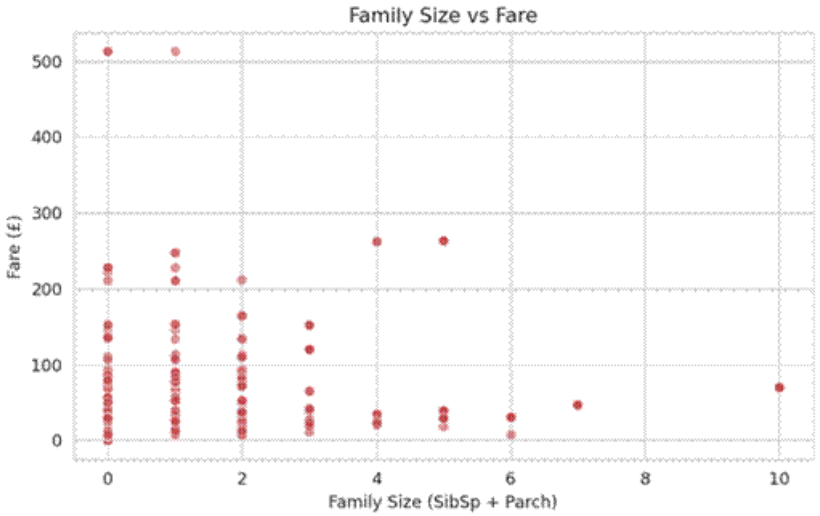



Figure 5(b): Family size vs Fare

Interpretation

- The relationship is weak and positive (r ≈ 0.22): passengers with family aboard tended to pay somewhat more.
- The majority (537 passengers, 60%) travelled alone, and their average fare was only £21.24, compared with £49.89 for those with one family member.
- Fare is not steadily rising with family size: it peaks around size 5 (average £73.72) and then falls for size 6 (£29.37), because large families were mostly in 3rd class.
- Note that the fare column can reflect the total ticket price for a group sharing one ticket, which may distort the relationship.

Q6. Create a countplot to show survival count by gender.

```
plt.figure(figsize=(6.5, 4))
```

```
ax = sns.countplot(x='Sex', hue='Survived', data=df, palette=['#C44E52',
'#55A868'])
for container in ax.containers:
        ax.bar_label(container)
plt.title('Survival Count by Gender')
plt.xlabel('Gender')
plt.ylabel('Number of Passengers')
plt.legend(title='Survived', labels=['No (0)', 'Yes (1)'])
plt.show()

print(df.groupby('Sex')['Survived'].mean() * 100)
```

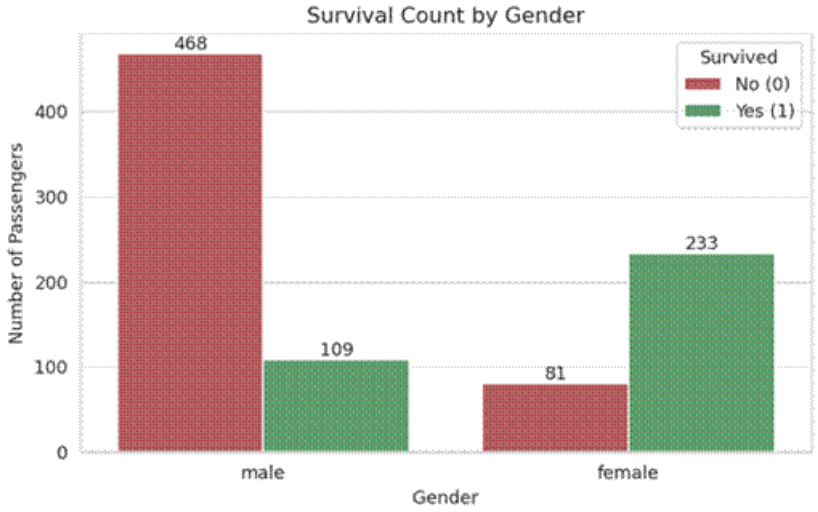



Figure 6: Survival count by gender

| Gender | Did Not Survive | Survived | Survival Rate |
|---|---|---|---|
| Female | 81 | 233 | 74.2% |
| Male | 468 | 109 | 18.9% |

Interpretation

- Gender had a dramatic effect on survival: 74.2% of women survived compared with only 18.9% of men.
- Although men made up almost two-thirds of the passengers, they account for the large majority of the deaths (468 of 549).
- This is consistent with the "women and children first" evacuation policy.

Q7. Create a bar chart to show survival by passenger class

```
survival_rate = df.groupby('Pclass')['Survived'].mean() * 100

plt.figure(figsize=(6.5, 4))
bars = plt.bar(['1st Class', '2nd Class', '3rd Class'], survival_rate.values,
                color=['#55A868', '#DD8452', '#C44E52'])
```

```
plt.bar_label(bars, fmt='%.1f%%')
plt.title('Survival Rate by Passenger Class')
plt.xlabel('Passenger Class')
plt.ylabel('Survival Rate (%)')
plt.ylim(0, 80)
plt.show()

print(pd.crosstab(df['Pclass'], df['Survived']))
```

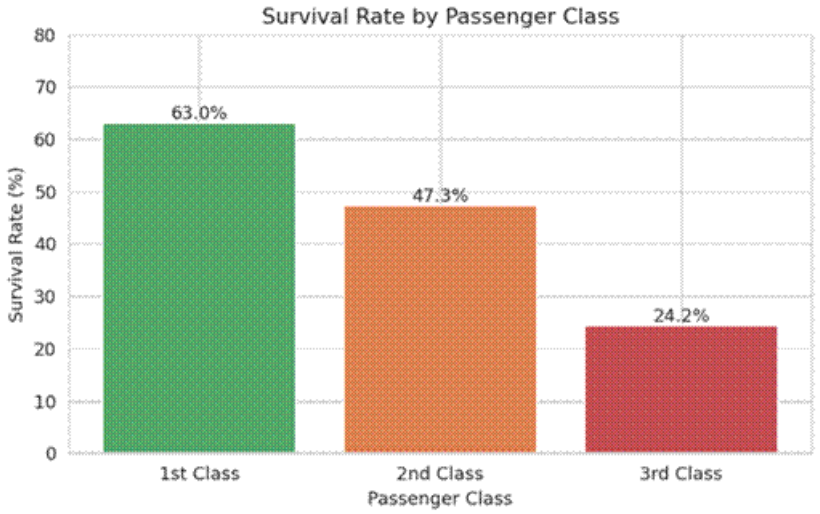



Figure 7: Survival rate by passenger class

| Passenger Class | Did Not Survive | Survived | Survival Rate |
|---|---|---|---|
| 1st Class | 80 | 136 | 63.0% |
| 2nd Class | 97 | 87 | 47.3% |
| 3rd Class | 372 | 119 | 24.2% |

Interpretation

- Survival falls steadily as class goes down: 63.0% (1st) → 47.3% (2nd) → 24.2% (3rd).
- A first-class passenger was about 2.6 times as likely to survive as a third-class passenger.
- In absolute numbers, 3rd class had the most deaths (372), even though it also had the most passengers. Better cabin locations and easier access to the lifeboats likely helped the upper classes.

Q8. Create a heatmap to show correlation between numerical variables.

```
num_cols = ['Survived', 'Pclass', 'Age', 'SibSp', 'Parch', 'Fare', 'FamilySize']
corr = df[num_cols].corr()

plt.figure(figsize=(7.5, 6))
```

```
sns.heatmap(corr, annot=True, fmt='.2f', cmap='coolwarm', center=0,
linewidths=0.5)
plt.title('Correlation Heatmap of Numerical Variables')
plt.show()
```

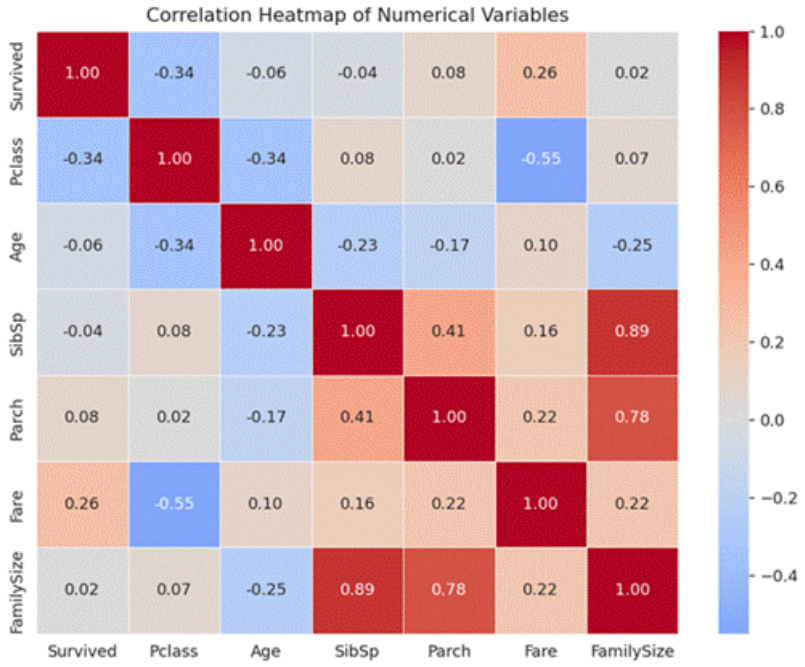



Figure 8: Correlation heatmap

Interpretation

- Pclass and Fare (−0.55): the strongest relationship in the data. A lower class number (1st) means a higher fare, which is expected.
- Survived and Pclass (−0.34): higher classes had better survival. Survived and Fare (0.26): paying more was linked to a better chance of survival, mostly because fare acts as a proxy for class.
- Age and Pclass (−0.34): older passengers tended to travel in higher classes.
- SibSp and Parch (0.41): passengers with siblings/spouses often also had parents/children aboard. FamilySize correlates strongly with both (0.89 and 0.78) because it is built from them, so it is redundant if all three are used in a model.
- Age, SibSp and FamilySize have almost no linear correlation with survival (between −0.06 and 0.02). Note that Sex is not shown, because it is a text column; it would need encoding to appear in the heatmap.

Q9. Identify which passenger class paid the highest fare on average

```
avg_fare = df.groupby('Pclass')['Fare'].mean().round(2)
print(avg_fare)
print('Highest average fare: Class', avg_fare.idxmax(), '->', avg_fare.max())
```

```
plt.figure(figsize=(6.5, 4))
bars = plt.bar(['1st Class', '2nd Class', '3rd Class'], avg_fare.values,
               color=['#55A868', '#DD8452', '#C44E52'])
plt.bar_label(bars, fmt='£%.2f')
plt.title('Average Fare by Passenger Class')
plt.xlabel('Passenger Class')
plt.ylabel('Average Fare (£)')
plt.show()
```

| Passenger Class | Average Fare |
|---|---|
| 1st Class | £84.15 |
| 2nd Class | £20.66 |
| 3rd Class | £13.68 |




Figure 9: Average fare by passenger class

Answer: 1st class paid the highest fare on average, at £84.15 per passenger.

Interpretation

- 1st-class passengers paid about 4 times the average 2nd-class fare and about 6 times the average 3rd-class fare.
- The 1st-class average is also pulled up by the extreme £512.33 tickets seen in the Q4 outliers.

Q10 Write at least 5 insights from the dataset based on your visualizations.

1. Gender was the strongest factor in survival. 74.2% of women survived versus 18.9% of men (Q6), despite men forming 64.8% of the passengers (Q1).
2. Passenger class strongly affected survival. Survival dropped from 63.0% in 1st class to 47.3% in 2nd and 24.2% in 3rd (Q7). Wealth and cabin location gave clear advantages.
3. More than half of the passengers were in 3rd class. 491 of 891 passengers (55.1%) travelled 3rd class (Q2), which explains why the overall survival rate was only 38.4% (342 of 891 survived).
4. Fares are highly skewed with many outliers. The median fare is £14.45 but the mean is £32.20; 116 passengers are outliers above £65.63, and three paid £512.33 (Q4).
5. 1st class paid far more on average. Average fare was £84.15 (1st), £20.66 (2nd) and £13.68 (3rd) (Q9), and Pclass–Fare is the strongest correlation (−0.55) in the heatmap (Q8).
6. Most passengers were young adults. Ages cluster between 20 and 40 (mean ≈ 29), with a small bump of young children and few passengers over 60 (Q3).
7. Age and family size are weak predictors of fare. Age vs fare correlation is only 0.10 and family size vs fare is 0.22 (Q5). Most passengers (537, i.e. 60%) travelled alone.
8. Fare and class carry the same information. Fare correlates positively with survival (0.26) mainly because it stands in for class, so both should not be treated as fully independent factors in modelling.

## 7. Conclusion

The EDA shows that survival on the Titanic was far from random. Gender and passenger class were the dominant factors: women and first-class passengers had much higher survival rates, while men and third-class passengers had the lowest. The passenger population was mostly young adults, mostly male and mostly third class, and ticket fares were heavily skewed by a small number of very expensive tickets. These findings would be a strong starting point for building a survival-prediction model, where Sex, Pclass and Fare should be the first features to include.